



Shane R. Spencer

Systems Architect & Mentoring Evangelist specializing in Linux, Python, and Cloud Ecosystems

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Leveraging 20+ years of systems expertise and a decade of SaaS architecture to build resilient, domain-centric platforms across high-growth startups and regulated healthcare/federal sectors. I prioritize **Developer Experience (DevEx)** and **Shift-Left** security to empower teams to accelerate time-to-market while neutralizing vulnerabilities and technical debt without operational friction.

CORE COMPETENCIES

AI-Driven Leadership & Agentic Workflows

- **Technical Judgment:** Strategic leverage of Large Language Models (LLMs) with a focus on prompt engineering and rigorous validation of AI outputs.
- **Agentic Engineering:** Advanced use of Claude, Cursor, and custom agentic loops to accelerate feature deliverability and enhance code clarity.
- **Architectural Automation:** Utilizing AI to automate repetitive DDD and Clean Architecture patterns, significantly reducing technical debt.
- **AI-Enhanced DevEx:** Integrating AI-powered guardrails and bespoke internal tooling directly into the developer workflow.
- **Agent & Tool Governance:** Implementing AI tools within regulated ecosystems (HIPAA/CMS) and focus on continuity and engineering practice requirements.

Software Architecture & Product Development

- **Languages:** Python), JavaScript/TypeScript, Shell, C/C++
- **Frontend & Mobile:** React, Redux, VueJS/NuxtJS, Vanilla HTML/CSS, RiotJS.
- **Design Patterns:** Clean Architecture, Domain-Driven Design (DDD), Event-Driven Architecture, Asynchronous Process Design, Microservice Prioritization.

Cloud Infrastructure & Platform Engineering

- **Cloud Platforms:** AWS (Advanced), Azure (General), Digital Ocean, CloudFlare.
- **Orchestration & DevOps:** Kubernetes, Docker, Serverless Framework, Terraform, SaltStack, Ansible, CI/CD (GitHub Actions, CircleCI, ArgoCD, Jenkins).
- **Edge & IoT:** Edge-level Caching, Site Synchronization, Linux Network Appliances, BMC Integrations, Command/Telemetry gathering.

Observability & Telemetry

- **Frameworks:** OpenTelemetry (Standardization & bespoke libraries), Prometheus, DataDog, AWS X-Ray, CloudWatch.
- **Network Monitoring:** SNMP Metric Generation, LogicMonitor, ThousandEyes, Distributed Tracing, Predictive Analytics, Real-time Alerting.

Security, Governance & Compliance

- **Security Mastery:** DevSecOps, Shift-Left Security, OWASP Benchmarks, Zero-Trust Architecture (BeyondCorp), Penetration Testing, Insider Threat Analysis.
- **Supply Chain Security:** Software Bill of Materials (SBOM) Automation, CycloneDX, SPDX, Dependency Scanning, Secure Boot (Signed Shim, iPX).
- **Regulatory Compliance:** HIPAA, HL7/FHIR, CMS (Medicare/Medicaid), FedRAMP, SOC I/II, VA/Federal PII & PHI Regulations.

Data Engineering & Specialized Systems

- **Database Management:** PostgreSQL (PostGIS), SnowflakeDB, MongoDB, DynamoDB, Redis, Intersystems IRIS/Caché, Oracle, PL/SQL.
- **Event Bus & Workflow:** AWS SQS, EventBridge, Kinesis, RabbitMQ, Celery, SMTP.
- **Specialized Domains:** Clinical Decision Support (CDS), Synthetic Data Generation (MITRE Synthea), GIS (PostGIS, Geographic Hashing, GDAL/OGR), Telecommunications (Asterisk PBX, SIP/IAX, Twilio).

WORK EXPERIENCE

07-2024

Document Engineering Manager / Full Stack Engineering Team Lead

ARINE

Document Engineering Manager: Directed a specialized team focused on developing scalable solutions for patient and provider letter generation utilizing technologies such as ReportLab and Typst. Architected novel document processing techniques that accelerated self-service letter production, eliminated developer bottlenecks, and improved overall feature time-to-market. This role encompasses the management of new software development for review management, mailroom/fax integration, and all CMS-regulated letter types. Leveraged MistQL and JSON Patch-based mutators to enable highly conditional outputs and point-in-time letter regeneration.

Document Automation | Developer Experience (DevEx) | Advanced Templating Engines | Feature Velocity

Full Stack Engineering Team Lead: Led backend and full-stack engineering teams in the architecture of secure, serverless SaaS portals (AWS Lambda, DynamoDB) supporting regulated clinical workflows and medication management. Ensured continuous compliance with CMS, Medicare, and Medicaid mandates while overseeing infrastructure and security within an AWS environment (SQS, API Gateway, Powertools). Maintained a strong emphasis on maintainability and observability across a stack spanning Python, JavaScript/TypeScript, React, and Redux.

Healthcare Compliance | Serverless Cloud Architecture | Cross-Functional Leadership | SDLC Governance

12-2020 - 07-2024

Strategic Technology Consultant & Senior Software Engineer

Various Organizations

Operated as a senior technical lead and subject matter expert for high-growth startups and large-scale enterprise ecosystems, delivering specialized solutions in observability, security, and cloud architecture.

Capital Group / Insight Global: Orchestrated the enterprise-wide transition of legacy network infrastructure monitoring to a resilient Prometheus and OpenTelemetry-based architecture, establishing a predictive observability fabric across extensive WAN links. Engineered automated Infrastructure-as-Code (IaC) workflows utilizing Ansible and bespoke Python developer tooling, drastically streamlining configuration management and elevating overall system reliability.

Enterprise Observability | Telemetry Standardization | Infrastructure-as-Code (IaC) / Custom Developer Tooling

Department of Veteran Affairs / DocMe360: Pioneered the integration of synthetic data generation systems to create privacy-compliant, Shift-Left testing environments, accelerating ML model validation while strictly adhering to HIPAA protocols. Governed secure, compliant infrastructure deployments within AWS GovCloud, architecting zero-trust data management frameworks that mitigated institutional risk for federal healthcare initiatives. Implemented centralized OpenTelemetry collectors to pre-filter data at the source, streamlining system auditing processes and concentrating federal compliance efforts into a unified collection point.

Synthetic Data Generation | Privacy-Preserving Architecture | Federal Compliance (GovCloud) / Centralized Telemetry

Adobe / TekSystems: Architected and deployed enterprise-grade Software Bill of Materials (SBOM) services utilizing Python and FastAPI, pioneering shift-left security strategies that proactively neutralized software supply chain vulnerabilities prior to deployment. Governed secure repository management and optimized Git processing engines across Adobe's massive global infrastructure, accelerating CI/CD pipelines via Kubernetes while strictly enforcing rigorous cybersecurity standards.

Software Supply Chain Security | SBOM Automation | CI/CD Governance | Enterprise Risk Mitigation

Serverless: Engineered bespoke OpenTelemetry Python libraries for serverless ecosystems, establishing asynchronous telemetry grouping that advanced system observability and accelerated new feature deployment for the core console product.

Telemetry Standardization | Asynchronous Architecture | Observability | Serverless Ecosystems

Taos / IBM Consulting: Contributed to enterprise architecture consulting initiatives, defining robust AWS data pipelines and integrating third-party platforms (Slack, Stripe) to accelerate outbound notification systems. Conducted rigorous dependency scanning and architectural reviews across 100+ microservices, enforcing SOC I/II and OWASP security benchmarks at scale while managing Python package development.

Enterprise Architecture Consulting | Microservices Governance | SOC I/II Compliance | Cloud Infrastructure

01-2023 - 05-2023

Principal System Engineer

Metify

Advanced platform-as-a-service (PaaS) provisioning solutions by engineering highly resilient Python web services and transitioning complex workloads to edge-optimized Debian/Ubuntu architectures.

Fortified the hardware-software boundary through the strategic implementation of Secure Boot protocols and custom key management, neutralizing supply chain threats across diverse x86 and ARM server deployments.

Edge Computing | Hardware-Level Security (Secure Boot) | Platform-as-a-Service (PaaS) | Continuous Integration/Deployment

12-2021 - 12-2022

Senior Software Development Engineer & Platform Engineer

EveryoneSocial / Gravit

Pioneered the strategic transition to highly scalable serverless architectures, eradicating infrastructure overhead and accelerating innovation velocity for a high-traffic SaaS platform. Architected custom SQS workflow engines and integrated AWS Athena data lakes to deliver predictive analytics and optimize data query performance across high-volume asynchronous environments.

Platform Engineering & Infrastructure: Architected and managed a transition to AWS serverless components including Lambda, DynamoDB, AppSync, and the Serverless Framework to support Fortune-500 client requirements. Optimized data warehousing and real-time operations by implementing AWS Timestream and Athena while refining AWS RedShift performance. Modified OpenTelemetry components for seamless integration with DataDog to ensure robust system observability and advanced user analytics.

Serverless Architecture | Data Warehousing | Observability | Infrastructure Optimization

Backend & Software Development: Designed a custom SQS workflow engine to streamline interactions with major social media platforms and ensure high-concurrency pipeline reliability. Developed and enhanced core product features using Python and React/React Native, with a heavy emphasis on GraphQL integrations and functional security. Established a comprehensive Software Bill of Materials (SBOM) using CycloneDX/SPDX to fortify telemetry and refine event validation across the product lifecycle.

High-Concurrency Pipelines | GraphQL Integrations | Python Packaging | Event-Driven Architecture

01-2002 - CURRENT

Technology Development & Technical Consulting

Brute Technologies / Sole Proprietor

Operating as a high-level technical strategist and builder for the startup ecosystem, specializing in translating complex business requirements into high-availability, cloud-native architectures. Directed end-to-end technology initiatives ranging from industrial IoT and advanced marketing analytics to specialized telemedicine and radiological data systems.

Telemedicine & Radiological Systems (Systems Specialist): Architected secure synchronization frameworks for HL7 and radiological information using AWS S3 to support clinical studies and remote imaging workflows. Engineered custom software for DICOM imagery processing in telemedicine applications, ensuring high-fidelity transmission and integration of custom health solutions for remote clinical work.

Remote Radiology | DICOM Processing | HL7/FHIR Standards | HIPAA Compliance

Gardyn / Ag-Tech (Strategic Lead): Engineered industrial IoT control systems for autonomous hydroponic environments, managing command-and-control telemetry for large-scale device fleets. Implemented Apache Kafka and cloud-native event buses to synchronize real-time sensor data and environmental telemetry gathered from devices in the field.

Industrial IoT | Apache Kafka | Hardware Integration | Real-Time Telemetry

Advertising & Media Technology (Analytics Architect): Architected high-volume data ingestion pipelines and real-time analytics dashboards using AWS Athena to optimize revenue reporting and spend integrity. Developed sophisticated post-processing frameworks for Video on Demand (VOD) and web advertisement projects focused on ad auditing and campaign management.

Big Data Analytics | AWS Athena | Revenue Integrity | High-Volume Ingestion

Telecommunications & Specialized Engineering (Lead Engineer): Developed high-capacity Asterisk-based telephony systems and distributed PBX architectures enhanced with machine learning components for increased functionality. Deployed specialized Linux implementations, including SnapCraft packaging and Ubuntu Core, on custom hardware for industrial applications.

Distributed Telephony | Asterisk PBX | Embedded Linux | Custom Tooling

08-2016 - 12-2020

Senior System Engineer

CGI / IBM / AT&T

Architected resilient, high-availability internal cloud infrastructures utilizing SaltStack and Ceph, enforcing Infrastructure-as-Code (IaC) principles to drive significant CapEx/OpEx efficiencies.

Implemented BeyondCorp-style zero-trust access controls and internal HTTP load balancers, ensuring continuous connectivity and rigorous regulatory adherence across a vast, geographically distributed telecommunications network.

Mission-Critical Infrastructure | Zero-Trust Access (BeyondCorp) | Infrastructure-as-Code (IaC) | High-Availability Systems

04-2013 - 01-2015

Geographic Information System (GIS) Specialist

ABR

Engineered custom GIS web applications to process and analyze massive terrestrial datasets (NASA/USGS), delivering critical business intelligence and optimizing field survey workflows.

Developed scalable PostgreSQL/PostGIS database architectures for large-scale imagery analysis, streamlining environmental research monitoring across geographically dispersed projects.

05-2008 - 07-2012

Complex Data Processing | Geospatial Architecture | Workflow Automation | Data Pipeline Engineering

● **Telecom Manager & Lead System Engineer**

Microcom / Sateo

Engineered distributed Asterisk PBX telephony systems and dynamic Python/Django web services to streamline field data capture and replace legacy operational workflows. Architected custom data integration frameworks with advanced signature processing, optimizing workforce monitoring and delivering actionable sales insights across a multi-state telecommunications network.

Telephony & Distributed Systems: Devised regional telephony services by intricately working with Asterisk PBX to bridge operations across Hawaii, Alaska, and Idaho. Developed custom modules and international voice termination software to manage call center operations and enhance inter-regional communication.

Distributed Systems | Asterisk PBX | Voice Operations | Legacy Integration

Web Services & Workflow Automation: Developed a custom Django-based web service for field data capture, replacing traditional methods and ensuring seamless integration with in-house systems. Engineered advanced signature processing—including detection, orientation, and alignment—to generate bespoke PDF reports with refined digital stamping. Refined ground-level data capture by reverse-engineering mobile-specific form input applications.

Workflow Automation | Django | Signature Processing | Operational Analytics